Understanding Clean-Energy Supply Chains: A US Perspective

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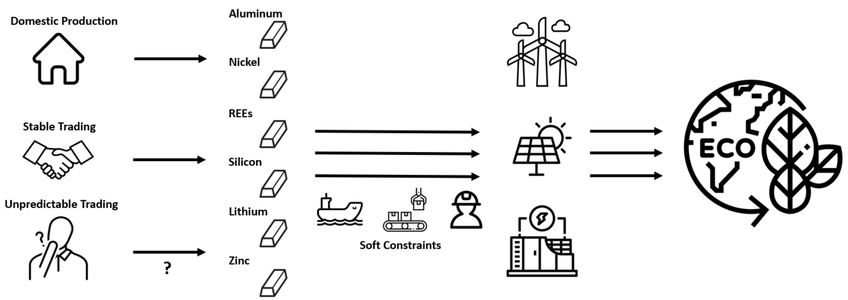
Introduction and Motivation

The clean energy transition is heavily dependent on the effective management of supply chains, particularly the availability of critical raw materials for manufacturing.



- Rising material demand creates bottlenecks, worsened by geopolitical tensions and concentrated sourcing.
- Existing studies lack comprehensive, nation-specific analyses of material supply chain constraints and are often prediction-based, lacking real-world applicability.

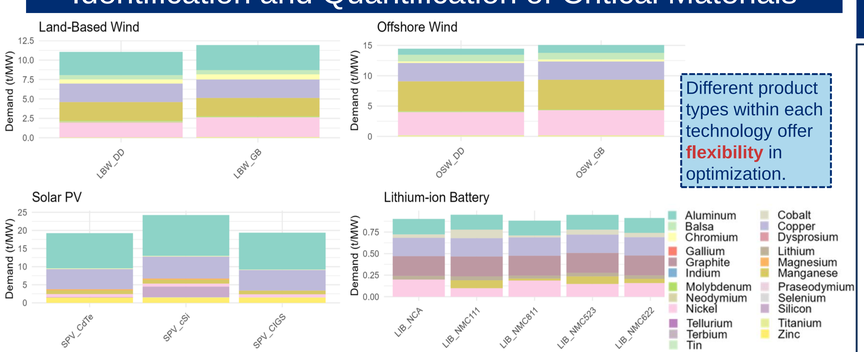
Three Key Questions (Progressively Deeper):

- ☑ **What critical materials and volumes are needed for wind, solar PV, and battery technologies?**
- ☑ **Can the US meet its net-zero milestones by optimizing available materials? What are the bottlenecks?**
- ☑ **How do geopolitical risks and non-collaboration scenarios impact clean energy transitions?**

Contributions

- **Material Identification and Quantification:** Identifies critical raw materials and maps their supply chain flows for wind, solar PV, and battery technologies.
- **Optimization Framework:** Creates a decision-making multi-annual optimization model to simulate clean energy planning under material and trade constraints through 2050.
- **Global Collaboration:** Examines the impact of trade partnerships and non-collaboration scenarios on the US clean energy transition.

Identification and Quantification of Critical Materials



Different product types within each technology offer **flexibility** in optimization.

Fig. 1: Raw material demand for clean energy products (aggregated from component-level)

Optimization Model Framework

Optimization Framework: Simulates clean energy deployment decisions over a multi-annual horizon with the following key features:

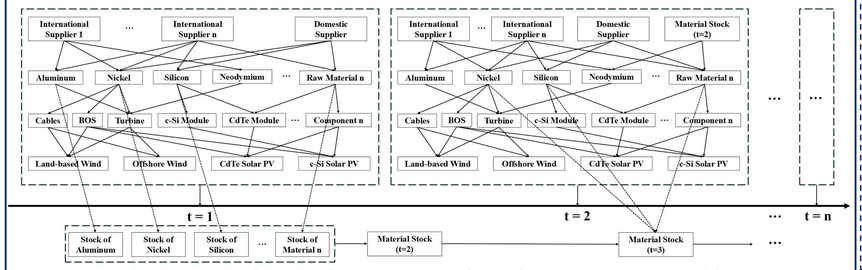


Fig. 2: Schematic of multi-annual for raw material supply and product assembly process

Objective (1):
Minimize total discounted costs for material procurement, production, and deployment.

$$\text{Min} \sum_{t\in\mathcal{T}} \frac{1}{(1+r)^{t-1}} \left(\sum_{m\in\mathcal{M}} C^{\text{rm}}_{m,t} \cdot \left(\sum_{o\in\mathcal{L}} p_{o,m,t} + \sum_{o\in\mathcal{D}} \frac{q_{o,m,t}}{R^{\text{c}}_{o,m,t}} + s_{m,t} \right) + \sum_{p\in\mathcal{P}} C^{\text{p}}_{p,t} \cdot x_{p,t} \right) \quad (1)$$

Supply Chain Process (2-3):
Ensure sufficient raw materials for components; sufficient components for products.

s.t.

$$y_{m,t} \geq \sum_{c\in\mathcal{C}} z_{c,t} \cdot D^{\text{c}}_{m,c}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T} \quad (2)$$

$$z_{c,t} \geq \sum_{p\in\mathcal{P}} x_{p,t} \cdot D^{\text{p}}_{c,p}, \quad \forall c \in \mathcal{C}, t \in \mathcal{T} \quad (3)$$

Raw Material Acquisition (4-6):
Ensure raw material procurement and domestic production stay within supplier limits; total usage remains within combined supply and stock levels.

$$y_{m,t} \leq R^{\text{d}}_{m,t} \left(\sum_{o\in\mathcal{L}} R^{\text{c}}_{o,m,t} \cdot p_{o,m,t} + \sum_{o\in\mathcal{D}} q_{o,m,t} \right) + R^{\text{s}}_{m,t} \cdot l_{m,t}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T} \quad (4)$$

$$p_{o,m,t} \leq Y_{o,m,t}, \quad \forall o \in \mathcal{L}, m \in \mathcal{M}, t \in \mathcal{T} \quad (5)$$

$$q_{o,m,t} \leq R^{\text{c}}_{o,m,t} \cdot Y_{o,m,t}, \quad \forall o \in \mathcal{D}, m \in \mathcal{M}, t \in \mathcal{T} \quad (6)$$

Stock Management (7-8):
Ensure stock levels are updated annually based on the balance of procurement, production, and consumption.

$$l_{m,t} = l_{m,t-1} + \left(\sum_{o\in\mathcal{L}} R^{\text{c}}_{o,m,t} \cdot p_{o,m,t-1} + \sum_{o\in\mathcal{D}} q_{o,m,t-1} - y_{m,t-1} \right), \quad \forall m \in \mathcal{M}, t \in \mathcal{T} \quad (7)$$

$$s_{m,t} \leq l_{m,t} - l_{m,t+1}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T} \quad (8)$$

$$\underline{X}^{\mathcal{K}}_t \leq \sum_{p\in\mathcal{K}} x_{p,t} \leq \overline{X}^{\mathcal{K}}_t, \quad \forall \mathcal{K} \in \{\mathcal{LW}, \mathcal{OW}, \mathcal{SP}, \mathcal{LB}\}, t \in \mathcal{T} \quad (9)$$

Deploy. Limits & Goals (9–10):
Ensure deployment stays within soft limits and meets cumulative goals over the planning horizon.

$$\sum_{p\in\mathcal{K}} \left(X_{p,0} + \sum_{\tau=1}^{t} x_{p,\tau} \right) \geq G^{\mathcal{K}}_t, \quad \forall \mathcal{K} \in \{\mathcal{LW}, \mathcal{OW}, \mathcal{SP}, \mathcal{LB}\}, t \in \mathcal{T} \quad (10)$$

$$p_{o,m,t} \geq 0,\ q_{o,m,t} \geq 0,\ s_{m,t} \geq 0,\ l_{m,t} \geq 0,\ x_{p,t} \geq 0,\ y_{m,t} \geq 0,\ z_{c,t} \geq 0, \quad \forall o, m, p, c, t \quad (11)$$

Non-negativity (11)

Importance of Global Collaboration and Trade

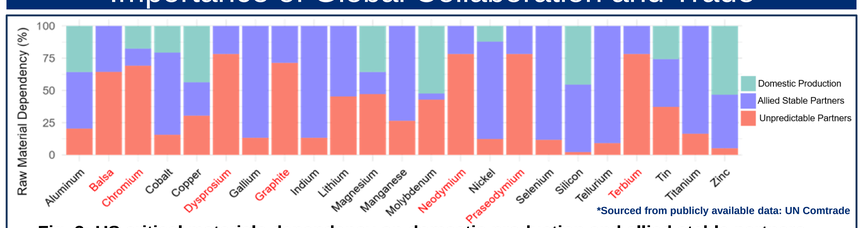


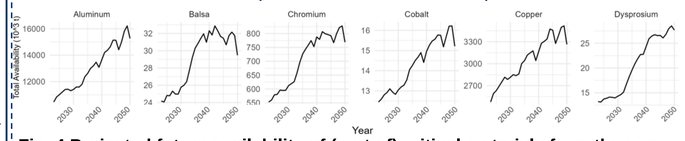
Fig. 3: US critical materials dependency on domestic production and allied stable partners

- Allied stable partners include **NATO, MNNA, AMNEA,** and Mexico under **NAFTA**.
- The US employs a **"friendshoring"** strategy, but some critical materials still rely heavily on imports from unpredictable partners.
- These include **balsa wood** (for turbine blades), **REEs**, and important **alloying materials**.

Case Study: Meeting US 2050 Net-Zero Goals

Model Inputs:

1. **Material demand schedule for identified clean energy technologies (Fig. 1)** ✚
2. **Projected future material availability based on the last supercycle* (Fig. 4)** ✚
3. **Regulated and government-set clean energy goals with soft limits.**
   - DOE targets
   - NREL studies
   - Net-zero roadmaps



***Supercycle**: Long-term price and demand trends driven by global technological advancements and resource consumption.

Fig. 4 Projected future availability of (part of) critical materials from the supercycle

Optimization

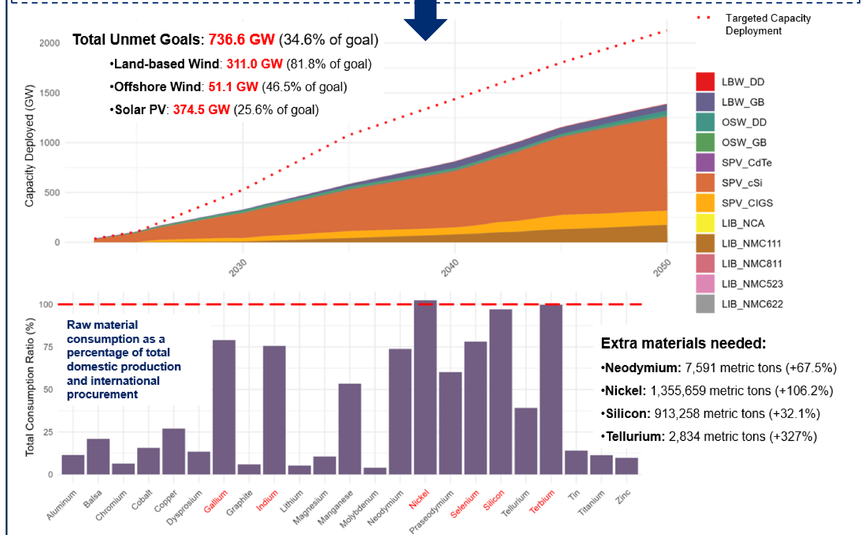


Fig. 5 Optimization results for clean energy product deployment and material consumption status

Case study on excluding unpredictable partners to reflect trade disruption scenario:

- Unmet goals increase by over **70 GW** across all technologies.
- More materials show scarcity, e.g., **silicon** and **REEs.**

Conclusion

- Critical materials like nickel, silicon, and REEs face severe shortages, limiting clean energy deployment.
- Current material accessibility is insufficient to meet 2050 clean energy goals, with over 1/3 of goals unmet under conservative estimates.
- The US heavily relies on imports, exposing supply chains to geopolitical risks.
- Solutions include boosting domestic production, recycling, material-efficient technologies, and strategic stockpiles.

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